

AUTOMATED WORKING LENGTH ESTIMATION FROM PERIAPICAL RADIOGRAPHS USING KEYPOINT DETECTION

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AUTOMATED KEYPOINT DETECTION ELIMINATES MANUAL MEASUREMENT IN ENDODONTICS

CLINICAL GOAL

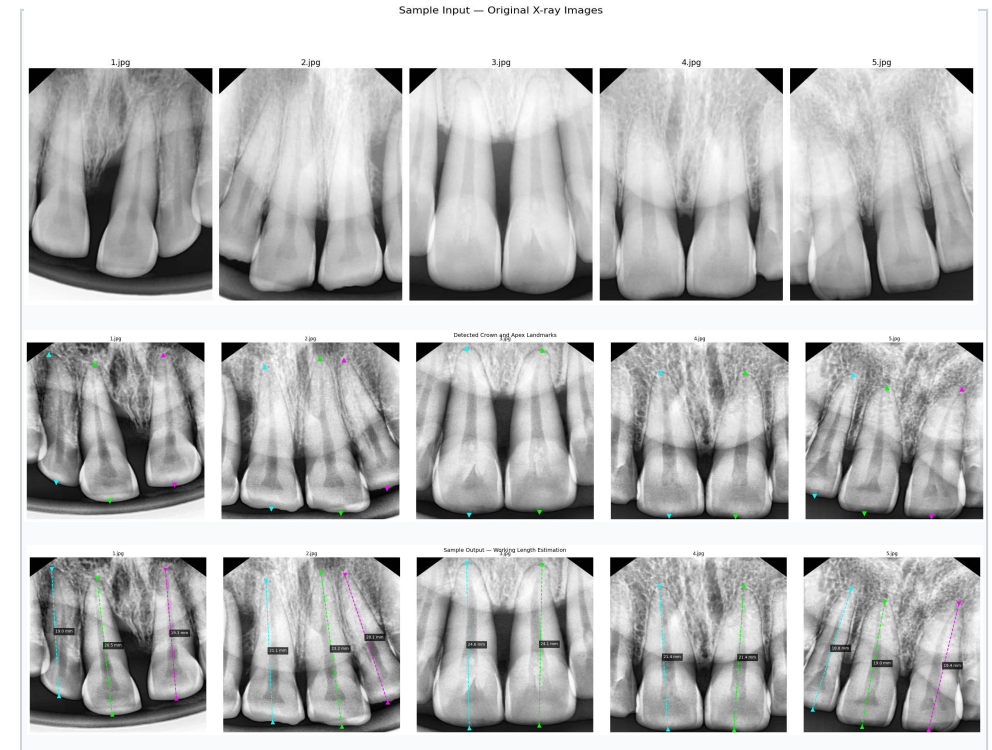
Develop a robust deep learning pipeline to automatically detect critical dental landmarks in radiographs.

TARGET LANDMARKS

Precision detection of both **crown and apex landmarks** in periapical dental X-rays.

AUTOMATION BENEFIT

Compute working length (WL) **without manual intervention**, reducing time and human error.



MULTI-STAGE UNET PIPELINE OPTIMIZES LANDMARK LOCALIZATION

CORE ARCHITECTURE

UNet + EfficientNet-B3

Utilizing a powerful encoder for high-performance feature extraction in medical imaging.

TRAINING STRATEGY

Pretraining & Fine-tuning

Pretrained on [DenPAR](#); fine-tuned on a custom teacher dataset of periapical X-rays.

HEATMAP PREDICTION

Dual Gaussian Heatmaps

Simultaneous prediction of heatmaps for both [crown tip](#) and [root apex](#) coordinates.

DATA AUGMENTATION

180° Rotation & Mirroring

Critical steps to handle raw images with apex at top, ensuring consistent model training.

ALGORITHMIC MATCHING ENSURES RELIABLE CROWN-APEX PAIRING

01

PEAK EXTRACTION

Landmark peaks are extracted from predicted heatmaps using **Connected Component Analysis** to identify precise coordinate centers.

02

LANDMARK MATCHING

The **Hungarian Algorithm** is employed to solve the bipartite matching problem, correctly pairing crowns and apexes in multi-tooth images.

03

CLINICAL CALIBRATION

Euclidean pixel distance is converted to physical units using a clinical calibration factor.

$$WL = (\text{Pixel Dist}) \times (21.5 \text{ mm} / 420 \text{ px})$$

PRELIMINARY EVALUATION CONFIRMS CLINICAL RANGE COMPLIANCE

13

PAIRS DETECTED

Total crown-apex pairs successfully identified across 5 test radiographs.

100%

RANGE COMPLIANCE

All predicted working lengths fell within the clinical 18–26 mm range.

X-ray image	Tooth	Working Length (mm)	Clinical status
Image 1	1	19.5	✓ Normal
	2	21.0	✓ Normal
	3	19.9	✓ Normal
Image 2	1	21.7	✓ Normal
	2	23.8	✓ Normal
	3	20.7	✓ Normal
Image 3	1	25.3	✓ Normal
	2	24.7	✓ Normal
Image 4	1	22.0	✓ Normal
	2	22.0	✓ Normal
Image 5	1	19.4	✓ Normal
	2	19.5	✓ Normal
	3	19.9	✓ Normal

Clinical range compliance used as proxy validation criterion in absence of ground truth.

AUTOMATED LOCALIZATION SUCCEEDS ACROSS DIVERSE TEST IMAGES

ZERO INTERVENTION

The model successfully localized all landmarks across all test images **without any manual intervention** or correction.

MULTI-TOOTH SUCCESS

Hungarian matching proved highly effective, correctly **pairing crowns and apexes** even in complex images containing multiple teeth.

ORIENTATION SENSITIVITY

The **180° flip augmentation** was critical; raw images often had the apex at the top, which was inconsistent with standard training orientation.

PLAUSIBLE ESTIMATES

Current working length (WL) estimates are **clinically plausible** and consistent with standard endodontic practice ranges.

QUANTITATIVE VALIDATION AND CANAL TRACING

DEFINE FUTURE WORK



QUANTITATIVE BENCHMARKING

Collect ground truth WL measurements from clinicians for **MAE and accuracy analysis** (within-2mm threshold).



DATASET EXPANSION

Increase the teacher dataset size beyond the initial 5 images to improve **model robustness** and generalization.



FULL PIPELINE INTEGRATION

Incorporate **segmentation-based canal tracing** for a comprehensive, end-to-end working length solution.

MODERN FRAMEWORKS SUPPORT HIGH-PRECISION MEDICAL IMAGING

DEEP LEARNING

PyTorch

Primary framework for model development and training.

SMP

[segmentation-models-pytorch](#) for EfficientNet-B3 UNet.

DATA & PROCESSING

OpenCV

Image manipulation and [CLAHE](#) contrast enhancement.

DenPAR

Public dental dataset used for model pretraining.

ALGORITHMS

Hungarian

Optimal bipartite matching for crown-apex pairing.

SciPy

Advanced peak detection and connected component analysis.